SWDiff: Stage-Wise Hyperspectral Diffusion Model for Hyperspectral Image Classification

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***Abstract*— Hyperspectral image classification (HSIC) has been a popular task in recent years. Even benefiting from the rapid development of deep neural networks (DNNs), there are still remaining intrinsic problems, including inadequate utilization of spatial–spectral information and insufficient labeled samples. The recent emergency of diffusion models (DMs) came to the fore because of their impressive refined image generation performance. DMs have been proven to not only can capture the underlying information of data through training the decoder of DMs, but also have more stable training than GANs while retaining even better performance. To better perceive and utilize spectral–spatial information while alleviating insufficient labeled samples simultaneously, we introduce the DM into HSIC from a data generation perspective. Specifically, we propose a stage-wise DM framework (SWDiff), dividing the HSIC task into three stages, including: pretrain the diffusion decoder with the hyperspectral image (HSI); generate new HSI cubes through the well-trained decoder to extra supply the original HSI set; and utilize the supplied dataset to train varied classifiers to obtain a better classification performance. Suitable pretraining could enable the decoder to acquire spatial–spectral information of the HSIs sufficiently via modeling spectral–spatial relationships across samples, leading to better utilization of spectral and spatial information of HSIs. Furthermore, the DM could provide the inference stage with spatial–spectral prior knowledge to ensure the feasibility and plausibility of the dataset complement, which could alleviate the insufficient labeled samples problem. Eventually, the classification stage will benefit from the first two stages.**

***Index Terms*— Classification, diffusion model (DM), hyperspectral images (HSIs).**

I. Introduction

HYPERSPECTRAL images (HSIs) consist of narrow wavelength bands at each pixel that contain high-resolution spectral features. Taking spectral and spatial information simultaneously into account, HSIs reveal more rich and unique information on land cover objects compared to RGB and multispectral (MS) data [1], [2]. Profited from this characteristic, HSIs have tremendous potential in basic but important tasks, such as classification, segmentation, and so on [3], [4], [5]. These crucial tasks can be extended to different high-level Earth observation (EO) missions including urban planning, mineral exploration, environment protection, and military [6].

Among the mentioned tasks, hyperspectral image classification (HSIC) is one of the most concerned tasks [2], [3]. Over the past decades, hand-crafted and learning-based feature extraction methods dominated the early period of the HSIC task, represented as random forest (RF), support vector machine (SVM), linear data analysis (LDA), principal component analysis (PCA), and so on. These methods achieve satisfactory classification accuracy in their own period. However, as the training dataset becomes more complex, they often reach a performance limit. This limitation is likely caused by the restricted data fitting, representation capabilities, and inadequate utilization of spatial and spectral information of these traditional approaches [7].

Fortunately, the rapid development of deep learning (DL) techniques empowers the better ability of representation and extraction features from HSIs due to the extraordinary capability of DL to extract meaningful, inherent, and potentially valuable knowledge from complex and pluralistic data. Motivated by the impressive achievements of DL techniques in computer vision (CV), numerous efforts have emerged to integrate DL frameworks and modules into HSIC. After several years of rapid development, as a result, plenty of widely recognized backbone networks have been successfully applied in the task of HSIC [8], such as 1-D and 2-D convolutional neural networks (1D/2D-CNNs) [9], [10], recurrent neural networks (RNNs) [11], [12], autoencoders (AEs) [13], [14], graph convolutional networks (GCNs) [1], and transformers [2]. These deep neural network (DNN)-based methods have acquired better capability of fitting and representing complex data, but not all these methods make the effort to utilize both the spatial and spectral information of HSIs, which leads to locally optimal performance. Furthermore, even though spectral–spatial feature extraction techniques amongst these DNNs-based methods (including CAE [15], stacked auto-encoders (SAEs) [14], spectralformer [2], and so on) have attempted to explore utilizing the spectral–spatial 3-D data characteristics, DNN-based methods remain suffer from inadequate utilization of spatial and spectral information due to their limitation to model spectral–spatial relationships across samples.

Apart from the perspective of feature extraction, suffering from insufficient labeled samples is another intrinsic and

Received 30 May 2024; revised 2 October 2024; accepted 18 October 2024. Date of publication 23 October 2024; date of current version 4 November 2024. This work was supported by the National Natural Science Foundation of China under Grant 62101041. *(Liang Chen and Jingfei He are co-first authors.) (Corresponding author: Hao Shi.)*

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Digital Object Identifier 10.1109/TGRS.2024.3485483

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common challenge in HSIC, limiting the performance of HSIC techniques. Combining generative models, which are well-known for their capability of fitting data distribution and generation samples, with the HSIC process is a feasible and intuitive approach to alleviate this problem [16]. Audebert et al. [17] utilized a GAN-based model to synthesize HSI examples from random distributions. The GAN conditioning ensures that the synthesized HSI examples correspond to the correct class. AC-WGAN-GP integrates generative models like GANs into HSIC processes to generate diverse and high-quality labeled data to address the challenge of insufficient labeled samples [18]. However, these GAN-based works confront the problem of unstable training, which extremely limits the employment and application of these methods.

Facing these intrinsic problems, inadequate utilization of spatial–spectral information and insufficient labeled samples, emerging diffusion models (DMs) provide new approaches to consider them. Recently, DMs have received significant success and widespread attention in generative tasks due to their refined generation performance and controllable generation process [19], [20].

For the former problem, inadequate utilization of spatial–spectral information, Baranchuk et al. [21] pointed out that DMs could be considered as meaningful and effective feature extractors to tackle not only generative tasks but also discriminative tasks, such as semantic segmentation. What's more, this idea empowers a novel paradigm to extract features with spatial and spectral information. Inspired by this thought, some works have been proposed to combine with HSIC tasks to improve the utilization of spatial–spectral information. SpectralDiff [22] and DiffSpectralNet [23] utilize DMs as feature extractors, whose discriminative capabilities vary with different timesteps. The DMs build the relationships between HSI samples so that the spatial–spectral features can be better extracted. Furthermore, Diff-HSI [24] introduces a feature dynamic fusion strategy to get spatial–spectral features for the classifier. Although these methods make efforts to introduce DMs into HSIC novelly, they all consider how to utilize DMs to tackle the intrinsic problems within HSIC from the discriminative standpoint, only focusing on obtaining better spatial–spectral features.

For the latter problem, insufficient labeled samples, DMs have demonstrated superiority to GANs in generation quality and training stability [25], so that data augmentation based on DMs will be more stable in training with retaining even better generation performance. For example, works like [26] prove DMs can outperform in data augmentation than GANs.

Inspired by Baranchuk et al. [21] and Chen et al. [26], we introduce DMs to better perceive and utilize spectral–spatial information while alleviating insufficient labeled samples simultaneously within the HSIC task. Specifically, we propose a stage-wise DM framework called SWDiff, which is more stable during the training process than GAN-based methods. This framework divides the HSIC task into three parts, including: pretrain the decoder of a DM with HSI cubes; generate new HSI cubes through the well-trained diffusion decoder to extra supply the original HSIC training set; and utilize the supplied dataset to train varied classifiers to obtain a better classification performance. Different from before mentioned diffusion-HSIC combining works concentrating on obtaining spatial–spectral features, the proposed SWDiff achieves better utilization of spatial–spectral information through a data generation approach. Suitable pretraining could enable the proposed DM decoder to acquire global spatial -spectral information of the HSIs sufficiently via modeling spectral–spatial relationships across samples. Consequently, the DM will be capable of perceiving and fitting the underlying data distribution of HSI, leading to better utilization of spectral and spatial information of HSIs. Furthermore, the DM could provide the generation stage with spectral and spatial prior knowledge to ensure the feasibility and plausibility of the dataset complement, which could alleviate the insufficient labeled samples problem. Eventually, the classification stage will benefit from the first two stages of this framework. The effectiveness of the proposed SWDiff framework has been proved through ample experiment results. The main contributions of this article can be summarized below.

1) To fully utilize spatial–spectral information and model spectral–spatial relationships across samples, we proposed SWDiff to accomplish conditional unsupervised generation via feature-map-level fusion within the denoising decoder guided by HSIs PCA components, which is compact but effective.
2) We proposed a lightweight decoder mainly comprised of the 2-D-convolution layer to settle the intrinsic disadvantage of the DM that revolved tremendous computational cost. This proposed decoder is at least 15 times lighter than diffusion milestone works, accelerating the sample speed eventually.
3) Benefiting from the lightweight decoder, once well pretrained, the proposed SWDiff could be considered as a plug-and-play component, which can be conveniently and quickly applied to even any HSIC methods from shallow to deep. Furthermore, this advantage enables the proposed SWDiff framework to be applied in wide scenarios, such as to be deployed on the FPGA board for professional utilization.
4) We conduct extensive experiments on three famous datasets (Indian Pines, Houston 2013, Augsburg), and the results show that our proposed stage-wise DM (SWDiff) effectively upgrades the classification performance of the current varied HSIs classification methods.

The rest of this article is organized as follows. Section II presents the related work including diffusion and HSIC. Section III elaborates on our proposed method, while Section IV details experimental results and further analysis. The last Section V summarizes this article and prospective work.

II. Related Work

In this section, we introduce the works related to our proposed framework, including the development of DMs, their applications in classification, and those in HSIC. The rapid evolution of DMs not only enables them with the potential to

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surpass other generative models, such as GANs, but also facilitates their applications in various traditional fields. Narrowing down to classification, especially HSIC, there are several novel works that have attempted to explore the combination of DMs with classification.

A. Diffusion Models

DMs, a class of generative models, have demonstrated significant potential in producing realistic samples [19], [20]. These models operate as parameterized Markov chains, trained via variational inference to generate samples conforming to data distributions within a finite sequence. DMs systematically degrade structured data, such as images, into unstructured noise and subsequently learn to restore the original data from this noise, generating samples that resemble the original.

Inspired by nonequilibrium thermodynamics, the prototype of DM was established in [27]. DM did not come to the fore until DDPM [28] was proposed. Unlike GANs, which can be challenging to train due to issues like mode collapse, DDPM [28] offered a more stable and theoretically grounded training process. This robustness stems from its foundation in stochastic differential equations and the gradual denoising process. Besides, DDPM [28] demonstrated an exceptional ability to generate high-quality and realistic samples, particularly in image generation, surpassing state-of-the-art generative models. Based on [28], DM began to draw widespread attention, leading to a surge of in-depth investigations and enhancements, such as DDIM [29], IDDPM [30], and stable diffusion [31].

These deviations are proposed from different aspects to improve the DDPM. For the exorbitant inference time cost, a notable drawback compared to other generative models [25], [29], [32], key advancements include accelerated sampling methods like DDIM [29], which simplifies inference by introducing implicit models. In addition, incorporating conditions such as text, categories, and features to guide the inference [25], [33], moves beyond the original unconditional generation paradigm [34]. Furthermore, traditional assumptions of Brownian motion in pixel space during diffusion have been reconsidered [28], [35]. Methods like [36] have redesigned latent spaces, and proposals exist for adopting non-Euclidean spaces—such as manifolds and graphs—for improved performance [37], [38].

Benefiting from the strengths and numerous improvements of DM, they have been widely utilized in various tasks of artificial intelligence. In the image generation conditioned on text, landmark works include Imagen [39], stable diffusion [31], and DALL-E·2 [40], while image-conditioned works are represented in [41] and [42]. In 3-D generation, [43], [44] directly produce 3-D point clouds, and [45] combines DM with NeRF. In video generation, representative works include Video DM [46], Make-A-Video [47], AnimatedDiff [48], FDM [49], and MCVD [50]. Furthermore, DM is also applied for text generation [51], time series generation [52], and audio generation [53], [54]. Apart from data generation tasks, DM is innovatively applied in other tasks, including super-resolution [55], image editing [56], inpainting [33], image segmentation [57], and medical analysis [58], among others.

B. DMs in Classification

Despite the significant success and widespread attention DMs have received in generative tasks, their application in classification tasks is still exploratory. The core questions include whether DMs can be applied to classification tasks and how they can assist in achieving better results. Some recent innovative works have provided interpretations to the aforementioned questions, summarized as follows.

Reference [59] is an early discussion about whether DMs can be used to perform classification, while they have had tremendous success in recent years. In this work, the tradeoff between the likelihood of the data and classification accuracy is deeply explored from the point of setting score-based generative models as classifiers. Besides, this work highlights that score-based generative models are closing the gap in classification accuracy compared to standard discriminative models. Another inspiring work on introducing DMs into classification tasks is [60]. It explores how to accurately predict a distribution based on a given conditional data distribution. Specifically, this work proposes classification and regression diffusion (CARD) models. CARD integrates covariate-dependence and a pretrained conditional mean estimator into both the forward and reverse diffusion chains of denoising diffusion probabilistic models. Essentially, it represents a class of conditional generative models that tackle supervised learning problems, implying that conditional generation is crucial for classification via DMs.

Based on conditional DMs, Subsequent works can be categorized into two branches: utilizing DMs for data-level [26] or feature-level [61] data augmentation to enhance classification accuracy; and considering DMs as classifiers, performing feature extraction and classification [62], [63].

Chen et al. [26] viewed DMs as a sound data-level augmentation approach, demonstrating that DMs can generate realistic and diverse synthetic images that effectively enhance classification accuracy. They proposed an environmental sound classification augmentation technique based on the diffusion probabilistic model with DPM-Solver++ for fast sampling. Differing from [26], [61] applies DMs to the enhancement of embedded medical images, which is feature map level augmentation instead of data-level. It introduces a diffusion-based model, DiffMIC, for medical image classification. DiffMIC is proposed to employ dual-granularity conditional guidance to process both global and local features of medical images, thereby enhancing classification accuracy in inference.

Apart from the perspective of data augmentation, DMs are also perceived as classifiers. Li et al. [62] emphasized that pretrained class-conditional DMs could be repurposed as classifiers without the necessity for additional training or fine-tuning, exploiting the potential of DM for zero-shot classification. Similar to [62], [63] attempts to use the feature maps extracted from the DM's decoder for classification. Specific feature selection and pooling ensure the increase of classification accuracy, as these intermediate feature maps

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contain discriminative information suitable for classification beyond noise prediction.

C. DMs in HSIC

In Section II-B, we categorize the introduction of DMs into classification into two branches: utilizing DMs for data augmentation to enhance classification accuracy and considering DMs as feature extractors to classify directly. In this subsection, we divide the exploring works of DMs in HSIC in line with this partition.

As previously mentioned, HSIC suffers from a lack of sufficient labeled samples, which can be mitigated through data augmentation. Currently, DM-based works in this branch are scarce, with only traditional DNN-based methods [64] and GAN-based works [17], [18] being utilized. In [17] and [18], a GAN-based model is utilized to conditional generate HSIs from random distributions to address the challenge of insufficient labeled samples. However, these methods are limited due to the inadequate generation ability and the unstable training caused by the model collapse of GANs.

Regarding the second branch, inspired by Baranchuk et al. [21], the primary paradigm is extracting feature maps with the noise of the DM decoder at a specific step t during training, then directly classifying using these feature maps. According to this paradigm, SpectralDiff [22] and DiffSpectralNet [23] utilize DMs as feature extractor layers without supplying the labeled samples. Besides, the discriminative capabilities vary with different timesteps. Furthermore, DiffHSI [24] introduces a feature dynamic fusion strategy to get HSI features for the classifier. Reference [65] also treats DM as a feature extractor to form an end-to-end HSIC framework. Reference [65] make progress by proposing an adaptively multitimestep multistage intermediate feature extracting module, CTMSFP, to extract multiple feature maps from varied t and stages. It is different from [22] and [23], which only extract a single feature map in a specific single timestep t from the mid layers of the DM decoder. Notably, this article innovatively considers the PCA of the HSI patches as the image to add noise, i.e., extracting feature maps from the PCA of HSI patches, rather than using the HSI patch itself, such as [22].

III. Methodology

In this section, the proposed SWDiff framework is described in detail. Through this framework, the original dataset is complemented through a diffusion decoder pretrained on the HSI dataset, ultimately leading to improved accuracy in HSIC via utilizing spatial–spectral information better and complementing labeled samples. The overall framework of the proposed SWDiff is illustrated in Fig. 1. This framework based on the DM is divided into three stages: Training, inference, and classification.

A. Training Stage of SWDiff Framework

In Fig. 1, the training stage corresponds to the left part. This stage primarily encompasses the PCA and cropping operation on HSI, forward diffusion processing, and reverse generation processing. Both the forward diffusion processing and reverse generation processing are closely involved with a conditional DM. Specifically, training this diffusion decoder could enable it to model spectral–spatial relationships across samples. Consequently, this well-trained decoder can sufficiently acquire global spatial–spectral information from the HSI, denoted as $\mathbf{HSI} \in \mathbb{R}^{B\times H\times W}$, where B, H, and W refer to the number of bands, height, and width of this HSI, respectively. Eventually, the utilization of spatial–spectral information can be improved, thereby enhancing both the generation quality and classification accuracy. Therefore, it is necessary to establish a proper bridge between HSIs and the DM.

DMs, essentially latent variable models comprising a Markov Chain with T steps, are employed to model the spectral–spatial relationships in HSIs. Rather than adding noise across the entire **HSI**—a theoretical but challenging approach due to limited training samples—we crop the HSI into uniform patches of size S. These patches, $\mathbf{HP}^{N\times(B\times S^2)}$, where N denotes the number of patches, B the number of bands, and S the patch size, are used to train the diffusion decoder.

The DM discussed above is unconditional, capturing spatial–spectral relationships within HSIs but only fitting the overall dataset distribution, not generating direction-specific images. To address it, condition setting—similar to feature pyramid networks (FPNs) in CV, typically through pretrained networks or blocks—is a primary solution. However, these pretrained modules are typically tailored to optical images, not HSIs, due to significant channel differences, creating a gap in suitable HSI-specific embeddings. PCA serves as a basic yet effective solution, facilitating the extraction and setting of PCA components as conditions for generation or inference. This approach not only enables conditional inference but also improves the utilization of spatial and spectral information while making further data complement possible. The Markov chain of DM could be reformulated as

$$p_\theta(\mathbf{HP}_0|\mathbf{P}) := \int p_\theta(\mathbf{HP}_{0:T}|\mathbf{P})d\mathbf{HP}_{1:T} \tag{1}$$

where $\mathbf{P}$ and $\mathbf{HP}_t$ are PCA components of corresponding $\mathbf{HP}_0$ and corrupted HSI patches with varying degrees of destruction depending on t, respectively. To elucidate the workflow of the SWDiff framework in the training stage clearly, two basic processing of DMs, forward diffusion processing and reverse generation processing, are needed to be formulized.

1) Forward Diffusion Processing: The forward diffusion processing is defined as gradually adding Gaussian noise into $\mathbf{HP}_0$ until $\mathbf{HP}_0$ completely destroyed into $\mathbf{HP}_T$, i.e., Gaussian noise too. Essentially, the forward diffusion processing splits this transformation into T pieces from $\mathbf{HP}_0$ to $\mathbf{HP}_T$, so that decomposing a complex transformation into a series of more small tasks. It gives the diffusion decoder the potential to capture spatial–spectral information underlying in $\mathbf{HP}_0$ in the next stage. The forward diffusion processing could be represented as

$$q(\mathbf{HP}_{1:T}|\mathbf{HP}_0) := \prod_{t=1}^{T} q(\mathbf{HP}_t|\mathbf{HP}_{t-1}). \tag{2}$$

Within the forward diffusion processing, noise selection is fundamental but key. In fact, any random distribution that can

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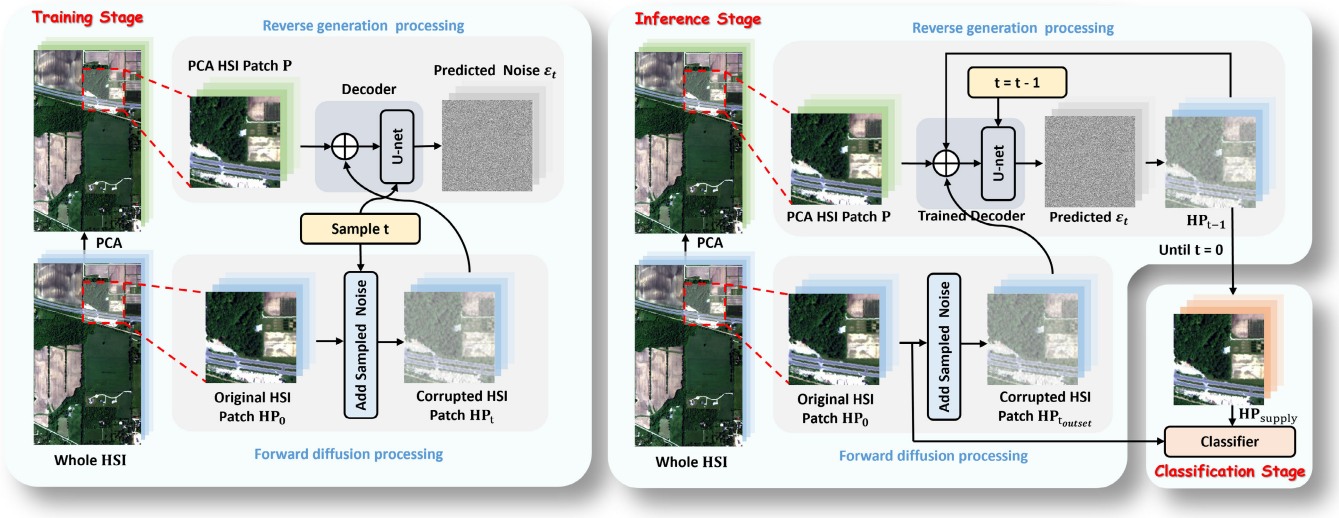


Fig. 1. Overview illustration of the proposed SWDiff framework for HSIC task. SWDiff comprises three parts including the training, inference, and classification stages.

be expressed as a function is theoretically capable. To simplify this part and in line with the DDPM, Gaussian distribution is our final choice which could be formulated as

$$\epsilon \sim \mathcal{N}(\epsilon; \mathbf{0}, \mathbf{I}). \tag{3}$$

Holding the same size as the $\mathbf{HP}_t$, the Gaussian noise ϵ is sampled individually in each diffusion step t according to a variance schedule $\beta_1, \ldots, \beta_T$. After we have selected a specific noise type, it is inefficient that we still need to sample and add noise on $\mathbf{HP}_0$ step by step to acquire the corrupted images $\mathbf{HP}_t$ with a certain step t. To solve it, with notating $\alpha_t := 1 - \beta_t$ and $\overline{\alpha}_t := \prod_{s=1}^{t} \alpha_s$, we utilize the reparameter skill to formulate $q(\mathbf{HP}_t|\mathbf{HP}_0)$

$$q(\mathbf{HP}_t|\mathbf{HP}_0) \sim \mathcal{N}\big(\mathbf{HP}_t; \sqrt{\overline{\alpha}_t}\mathbf{HP}_0, (1 - \overline{\alpha}_t)\mathbf{I}\big). \tag{4}$$

It is notable that the variance schedule has not only one option. The common two noise scheduling strategies are linear and cosine. We hope to destroy the original $\mathbf{HP}_0$ progressively and not completely lose all the information of $\mathbf{HP}_0$ until the last step. However, the linear schedule is prone to destroy the $\mathbf{HP}_0$ too fast in the late step, such as the 950 step of $T = 1000$. It will lead to nearly entire information being lost earlier than the last step T. Consequently, the last many steps will be redundant and adverse to the diffusion decoder to capture the spatial–spectral information within $\mathbf{HP}_0$. To put it another way, these steps will make no contribution to the training even be harmful. On the other hand, the phenomenon of early loss of information will also elevate the difficulty of predicting the noise of the final steps, which might decrease the inference quality in the next stage. Fortunately, the cosine schedule can mitigate this issue, as within the strategy of the cosine schedule, even though the value of β_t increases with each step t, the rate of increase in β_t progressively slows down. It will ensure the diffusion decoder stably captures refined spatial–spectral information and model spectral–spatial relationships across samples.

2) Reverse Generation Processing: The reverse generation processing is defined as predicting the noise ϵ_t at every randomly sampled step t ranging from 0 to T. Although the reverse generation is not directly executed during the training stage, the derivation of the training algorithm is based on the definition of the reverse generation process. To put it another way, each denoising step in the training process simulates a separate step in the reverse process, though a complete sampling from noise to data does not occur during actual training. Therefore, this process is an essential component of the training model. Similar to forward diffusion processing, reverse generation processing can also be formulated as a joint distribution

$$p_\theta(\mathbf{HP}_{0:T}|\mathbf{P}) := p(\mathbf{HP}_T) \prod_{t=1}^{T} p_\theta(\mathbf{HP}_{t-1}|\mathbf{HP}_t, \mathbf{P}) \tag{5}$$

where θ represents the parameters that need to be predicted through a network. In contrast, the previously mentioned forward diffusion processing does not include parameters to be predicted. This formula is further defined as a Markov chain with learned Gaussian transitions starting at $p(\mathbf{HP}_T) = \mathcal{N}(\mathbf{HP}_T; \mathbf{0}, \mathbf{I})$

$$p_\theta(\mathbf{HP}_{t-1}|\mathbf{HP}_t, \mathbf{P}) := \mathcal{N}(\mathbf{HP}_{t-1}; \mu_\theta(\mathbf{HP}_t, \mathbf{P}, t), \Sigma(\mathbf{HP}_t, \mathbf{P}, t)) \tag{6}$$

where $\Sigma(\mathbf{HP}_t, \mathbf{P}, t)$ is set as $\beta_t\mathbf{I}$ and learned parameters are contained in $\mu_\theta(\mathbf{HP}_t, \mathbf{P}, t)$. $\mu_\theta(\mathbf{HP}_t, \mathbf{P}, t)$ could be represented as

$$\mu_\theta(\mathbf{HP}_t, \mathbf{P}, t) = \frac{1}{\sqrt{\alpha_t}}\left(\mathbf{HP}_t - \frac{\beta_t}{\sqrt{1 - \overline{\alpha}_t}}\epsilon_\theta(\mathbf{HP}_t, \mathbf{P}, t)\right) \tag{7}$$

where ϵ_θ is a function approximation intended to predict ϵ_t from $\mathbf{HP}_t$. During the reverse generation processing, the corrupted $\mathbf{HP}_t$, along with specific step t, and corresponding PCA components $\mathbf{P}$ are input together into the diffusion denoising decoder. Besides before mentioned ϵ_{t-1} can be set as the out of the diffusion decoder to construct the bridge between

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HSIs and DMs, $\mathbf{HP}_{t-1}$ and μ_{t-1} of the former step in forward diffusion processing are also could be selected as the output to predict. According to DDPM [28], directly predicting the former step image $\mathbf{HP}_{t-1}$ could lead to unstable training and lower quality of the final generated images than predict other parameters. They also found that predicting the noise ϵ_{t-1} outperformed predicting the mean μ_{t-1}. Ultimately, we chose to predict the noise ϵ_{t-1}; in other words, we set the output of the decoder to ϵ_θ. It is not only because it resulted in the best performance in our experiments but also because predicting ϵ_{t-1} simplifies the model's training to some degree. To sum up, the algorithm of the training stage is shown below.

Algorithm 1 Training

1: **repeat**
2: $\mathbf{HP}_0 \sim q(\mathbf{HP}_0)$
3: $t \sim \text{Uniform}(\{1, \ldots, T\})$
4: $\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
5: Take gradient descent step on
$$\nabla_\theta \|\epsilon - \epsilon_\theta(\sqrt{\overline{\alpha}_t}\mathbf{HP}_0 + \sqrt{1-\overline{\alpha}_t}\epsilon, t|\mathbf{P})\|^2$$
6: **until** converged

B. Inference Stage of SWDiff Framework

In Fig. 1, the inference stage of the proposed SWDiff framework corresponds to the right upper part. This stage is defined as sequentially predicting the noise ϵ_t through a well-trained diffusion decoder and subtracting ϵ_t from $\mathbf{HP}_t$ at each step t ranging from T to 0. Specifically, after multiple training iterations in the former training stage, the diffusion decoder eventually becomes capable of generating an HSI patch conditioned on an original patch by sequentially predicting and denoising the noise, with the utilization of the spatial–spectral information captured by the diffusion decoder.

The key to generation in a certain direction is the conditional mechanism. Besides the PCA component $\mathbf{P}$ discussed before, the setting of the generation outset of step t is also important for providing guidance information. When corrupted $\mathbf{HP}_t$ is set as the outset instead of $\mathbf{HP}_T$, an isotropic Gaussian noise, $\mathbf{HP}_t$ itself will provide the retaining information that did not be destructed yet as conditional information. Moreover, smaller t equals more retaining information, leading to more similar generation results $\mathbf{HP}_{\text{supply}}$ compared with the original image patch $\mathbf{HP}_0$.

DDIM [29], a crucial improvement of DDPM, reconstructed DDPM's sampling stage, leading to accelerated sampling. However, in our framework, we adopted a sampling method similar to DDPM's rather than DDIM. This is because, thanks to the lightweight network design, satisfactory sampling speeds are achieved even without the use of the DDIM accelerated sampling algorithm. With notating the outset step of generation as t_{outset}, the overall algorithm of the inference stage is shown below.

C. Classification Stage of SWDiff Framework

In the third SWDiff classification stage, the HSI classifier is trained using both the original HSI patches $\mathbf{HP}$ and the newly generated complemented patches $\mathbf{HP}_{\text{supply}}$ from the inference stage. To display the effectiveness of the proposed SWDiff, the classification networks are intended to be in line with the corresponding original papers. The setting and analysis of classification will be discussed in detail later in Section IV.

Algorithm 2 Inference

1: $\mathbf{HP}_{t_{outset}} \sim q(\mathbf{HP}_{t_{outset}}|\mathbf{HP}_0)$
2: **for** $t = t_{outset}, \ldots, 1$ **do**
3: $\mathbf{z} \sim \mathcal{N}(0, I)$ if $t > 1$, else $\mathbf{z} = 0$
4: $\mathbf{HP}_{t-1} = \frac{1}{\sqrt{\alpha_t}}(\mathbf{HP}_t - \frac{1-\alpha_t}{\sqrt{1-\overline{\alpha}_t}}\epsilon_\theta(\mathbf{HP}_t, t|\mathbf{P})) + \sigma_t\mathbf{z}$
5: **end for**
6: **return** $\mathbf{HP}_{supply}$

D. Loss Function

As discussed before, ϵ_t is the final choice of the decoder in SWDiff to predict. The loss function of the training stage starts from maximizing the likelihood. Combined with a series of simplifications, the loss function of SWDiff's training Stage is eventually a mean-square error (mse), represented as

$$L_{\text{simple}}(\theta) := \mathbb{E}_{t,\mathbf{HP}_0,\mathbf{P},\epsilon}\left[\left\|\epsilon - \epsilon_\theta\left(\sqrt{\overline{\alpha}_t}\mathbf{HP}_0 + \sqrt{1-\overline{\alpha}_t}\epsilon, \mathbf{P}, t\right)\right\|^2\right]. \quad (8)$$

By restricting the output of the decoder to be similar to the corresponding true noise ϵ_t, the diffusion decoder will gradually be aware of the underlying spatial–spectral information relationship across samples. In addition, the function of the classification stage is set to cross-entropy (CE).

E. Lightweight Designment of Decoder

A critical limitation of DMs is their high computation cost and time consumption in the inference stage. Accelerating inference algorithm is one of the solutions exemplified by DDIM [29] that has been proposed to hasten the sampling process of DDPM [28]. Apart from this kind of method, the proposed SWDiff is intended to lighten the decoder model while retaining the advantages of DMs, which can also reduce the computational overhead and duration of inference.

The network architecture is depicted in Fig. 2. In contrast to other DDPM-based decoders, this decoder network adheres to the basic structure of U-Net while simplifying the feature extraction layers and time embedding layers. Regarding the condition, The PCA component $\mathbf{P}$ is fused with the feature map of $\mathbf{HP}_t$ via add to provide indispensable guidance during conditional generation.

IV. Experiments

To validate the proposed SWDiff's performance in quantity and quality, comprehensive experiments have been conducted on three different datasets, including Indian Pines, Houston 2013, and Augsburg. The experiments primarily focused on comparing the classification performance changes of existing stem HSIC methods before and after using SWDiff.

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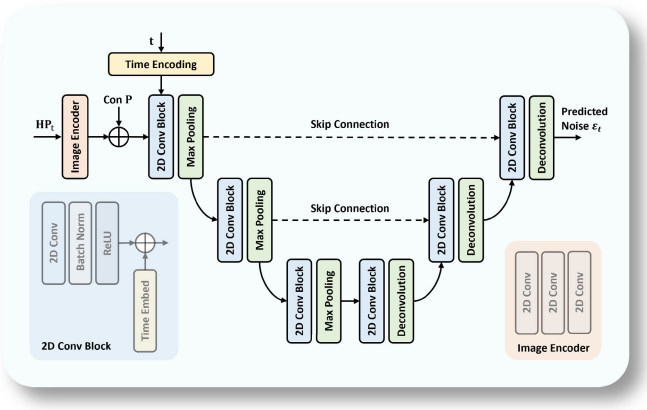


Fig. 2. Architecture of the denoising decoder in training and inference stages.

TABLE I

INDIAN PINES DATASET: THE NUMBER OF TRAINING AND TEST SAMPLES AND THE TOTAL NUMBER OF SAMPLES PER CLASS

Class No.	Class Name	Training	Test	Total
1	Corn Notill	50	1,384	1,434
2	Corn Mintill	50	784	834
3	Corn	50	184	234
4	Grass Pature	50	447	497
5	Grass Trees	50	697	747
6	Hay Windrowed	50	439	489
7	Soybean Notill	50	918	968
8	Soybean Mintill	50	2,418	2,468
9	Soybean Clean	50	564	614
10	Wheat	50	162	212
11	Woods	50	1,244	1,294
12	Buildings Grass Trees Drives	50	330	380
13	Stone Steel Towers	50	45	95
14	Alfalfa	15	39	54
15	Grass Past Mowed	15	11	26
16	Oats	15	5	20
Total		695	9,671	10,366

A. Data Description

1) Indian Pines Dataset: The Indian Pines dataset, procured using the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) in 1992, is a cornerstone in HSI analysis. The dataset covers an agricultural area in Northwestern Indiana, consisting of 16 classes representing different crop types and natural vegetation. These classes include corn, soybean, residues, and wheat. With 220 spectral reflectance bands in the wavelength range from 400 to 2500 nm and a spatial resolution of 20 m per pixel. Notably, we did not remove the 20 noisy and water absorption bands within 220 as other HSIC work did, such as 104–108, 220, and the like. The ground truth contains 10 366 labeled pixels, divided into 695 in the train set and 9671 in the test set. Table I depicts the dataset in all classes. The details and division of each category of Indian Pines are shown in Table I. We draw the false-color image of the Indian Pines dataset in Fig. 3.

2) Houston 2013 Dataset: Houston 2013 dataset was introduced for the 2013 IEEE Geoscience and Remote Sensing Society (GRSS) Data Fusion Contest. This dataset includes HSIs taken over the University of Houston campus and the neighboring urban area in TX, USA. With 144 spectral bands

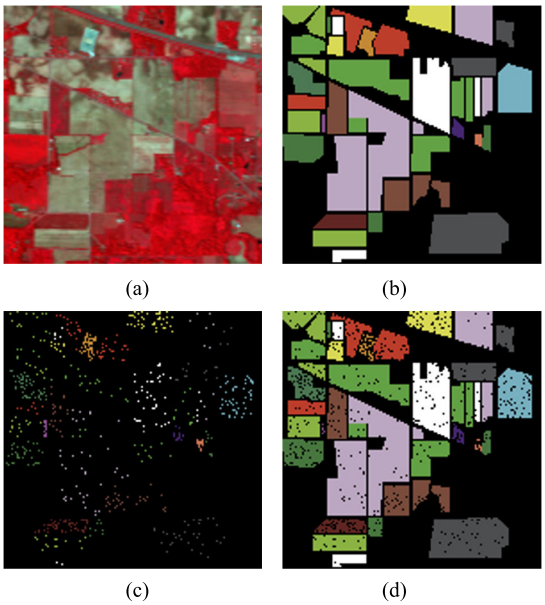


Fig. 3. Samples of the Indian Pines dataset. (a) False-color image (bands 57, 27, and 17). (b) Ground truth. (c) Training samples. (d) Test samples.

TABLE II

HOUSTON 2013 DATASET: THE NUMBER OF TRAINING AND TEST SAMPLES AND THE TOTAL NUMBER OF SAMPLES PER CLASS

Class No.	Class Name	Training	Test	Total
1	Healthy Grass	198	1,053	1,251
2	Stressed Grass	190	1,064	1,254
3	Synthetic Grass	192	505	697
4	Tress	188	1,056	1,244
5	Soil	186	1,056	1,242
6	Water	182	143	325
7	Residential	196	1,072	1,268
8	Commercial	191	1,053	1,244
9	Road	193	1,059	1,252
10	Highway	191	1,036	1,227
11	Railway	181	1,054	1,235
12	Parking Lot 1	192	1,041	1,233
13	Parking Lot 2	184	285	469
14	Tennis Court	181	247	428
15	Running Track	187	473	660
Total		2,832	12,197	15,029

ranging from 364 to 1046 nm, the whole scene is made of 349 × 1905 pixels and at 10-nm spectral resolution. The dataset includes 15 land cover classes for classification tasks, including grass, trees, soil, water, residential, commercial, road, highway, railway, parking lot, tennis court, and running track, among other subclass. Table II lists these 15 classes in detail, also involving the partition of the whole dataset from the train set to the test set, which contains 2832 and 12 197 pixels, respectively. We draw the false-color image of the Houston 2013 dataset in Fig. 4.

3) Augsburg Dataset: The Augsburg dataset represents an urban-centric hyperspectral collection, offering a detailed spectral snapshot of Augsburg, Germany. This scene includes three data sources: an HSI, a dual-pol synthetic aperture radar (SAR) image, and a digital surface model (DSM). In this article, only the HSI was used. With 332 × 485 pixels in HSI, the Augsburg dataset comprises 180 spectral bands ranging

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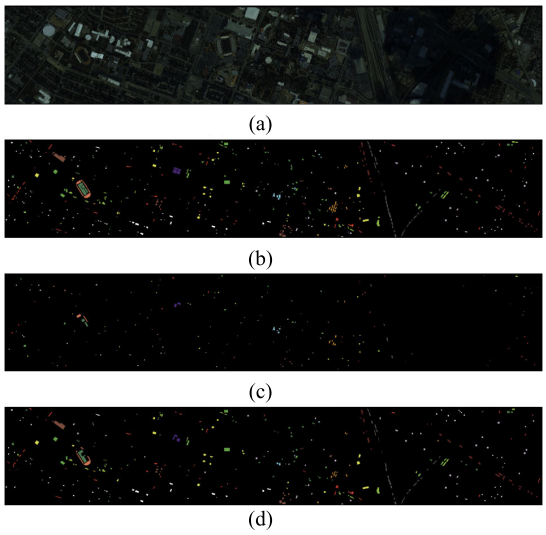


Fig. 4. Samples of the Houston 2013 dataset. (a) False-color image (bands 70, 50, and 20). (b) Ground truth. (c) Training samples. (d) Test samples.

TABLE III
AUGSBURG DATASET: THE NUMBER OF TRAINING AND TEST SAMPLES AND THE TOTAL NUMBER OF SAMPLES PER CLASS

Class No.	Class Name	Training	Test	Total
1	Forest	146	13,361	13,507
2	Residential Area	264	30,065	30,329
3	Industrial Area	21	3,830	3,851
4	Low Plants	248	26,609	26,857
5	Allotment	52	523	575
6	Commercial Area	7	1,638	1,645
7	Water	23	1,507	1,530
Total		761	77,533	78,294

from 400 to 2500 nm. It delineates seven unique urban land cover classes, such as forest, water, residential, and industrial areas, among others. This dataset assigns 761 pixels to the train set and 77 533 to the test set. Table III details the partition between the train and test sets of these classes. We draw the false-color image of the Augsburg dataset in Fig. 5.

B. Experimental Setting

A variety of methods used in HSIC were employed to validate the effectiveness of the proposed SWDiff. These methods include RF, 2DCNN, CAE, SSRN, and AMS-M2ESL [66]. Specifically, the classification performance of these networks was compared with and without the application of the SWDiff. The selection of methods encompassing both machine learning and DL aims to demonstrate the generality of SWDiff thoroughly. Moreover, in the classification stage, all experiments were conducted (including those with or without applied proposed SWDiff) under the same experimental conditions and hyperparameter settings to ensure a fair comparison.

Three datasets were divided into training and test sets, with the details of the division illustrated in Tables I–III. In these datasets, both Indian Pines and Augsburg contain categories with a limited number of training samples. For instance, in Indian Pines, classes such as Alfalfa, Grass Past Mowed, and Oats each have only 15 training samples; in Augsburg, classes Industrial Area, Commercial Area, and

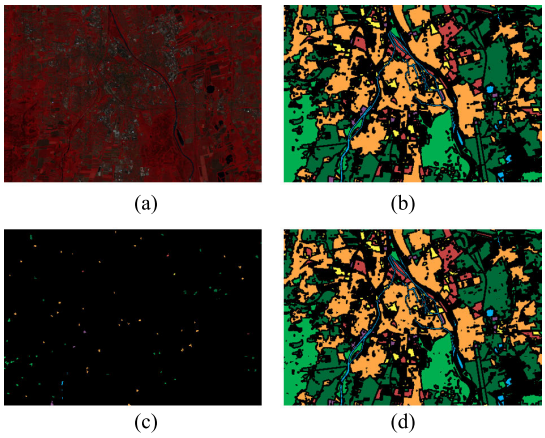


Fig. 5. Samples of the Augsburg dataset. (a) False-color image (bands 40, 20, and 10). (b) Ground truth. (c) Training samples. (d) Test samples.

Water have only 2, 7, and 23 samples, respectively. Besides, all three datasets exhibit unbalanced labels, especially in Indian Pines and Augsburg. Specifically, in Indian Pines, the class with the most labels, Soybean Mintill, has over 100 times more labels than Oats, while in the latter two datasets, the difference is approximately 4 and 52 times, respectively. Such significant disparities in label quantities between different classes within the same dataset present considerable challenges and difficulties for classification.

Hyperparameters in our study can be broadly categorized into those related to the DM and those pertinent to classification. The hyperparameters associated with the DM primarily include the range of β, the end of timestep T, the initial timestep for generation t_{outset} and the number of samples n_{sample} generated for each HSI patch **HP** during the dataset cropping stage. The range of β is set as $[0.0001, 0.02]$, which is involved with the noise level in each t. Under the given variance schedule, the end of step T in the proposed method is set to 1000, consistent with the settings in the DDPM [28]. Setting T to 1000 ensures that the **HP** are ultimately noised into isotropic Gaussian noise while avoiding the excessive t that could lead to slow training and inference. The settings and impacts of t_{outset} and n_{sample} will be discussed later. The learning rate for the training stage is set to $2e^{-5}$, aligning with the settings used in the DDPM. The number of PCA components is set to 3. As for the hyperparameters related to classification, the principle is to ensure fairness in experimental comparisons. In all experiments involving the training of classifiers, the batch size and patch size are set to 64 and 16, respectively, with a learning rate of 0.01.

Similar to other works in HSIC, to evaluate the impact of SWDiff on classifiers, overall accuracy (OA), average accuracy (AA), and the kappa coefficient (Kappa) are used as evaluation criteria. The experimental results for the three datasets are the average of eight separate runs. In addition, when employing SWDiff, multiple classifiers within one dataset utilize the same trained diffusion decoder, which is trained during the first stage of the experiment. The experiments are executed in Python 3.8.17 and PyTorch 2.0.1, with the hardware of NVIDIA A6000 GPU and Intel[1] Xeon[1] Platinum 8336C CPU.

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TABLE IV

CLASSIFICATION RESULTS BY DIFFERENT METHODS WITH OR WITHOUT SWDIFF FOR THE INDIAN PINES DATASET, INCLUDING PER-CLASS ACCURACY, OA, AA, AND KAPPA

	RF		CNN		CAE		SSRN		AMS-M2ESL	
Class No.	Before	After	Before	After	Before	After	Before	After	Before	After
1	62.38 ± 1.03	**65.75 ± 1.05**	90.93 ± 1.93	**92.23 ± 1.79**	91.74 ± 1.11	91.51 ± 1.18	91.92 ± 2.34	**92.99 ± 1.65**	96.56 ± 1.15	**97.49 ± 0.52**
2	86.89 ± 1.13	**90.66 ± 0.48**	97.51 ± 1.04	**97.86 ± 1.11**	98.28 ± 1.20	97.80 ± 0.72	97.91 ± 0.87	**98.37 ± 0.85**	96.21 ± 2.23	**96.94 ± 0.65**
3	87.02 ± 1.06	**88.59 ± 1.16**	99.80 ± 0.40	**99.87 ± 0.25**	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00	97.36 ± 1.93	94.64 ± 1.10
4	94.55 ± 0.29	**96.31 ± 0.55**	94.86 ± 0.57	**95.64 ± 0.96**	94.47 ± 1.28	**95.36 ± 0.74**	94.41 ± 1.95	**96.24 ± 2.07**	97.58 ± 1.88	**97.96 ± 0.50**
5	90.64 ± 0.45	**91.86 ± 0.72**	99.95 ± 0.07	99.88 ± 0.09	99.71 ± 0.20	100.00 ± 0.00	98.28 ± 1.19	**99.60 ± 0.25**	99.09 ± 0.75	**99.35 ± 0.33**
6	100.00 ± 0.00	100.00 ± 0.00	99.80 ± 0.08	**99.83 ± 0.11**	99.94 ± 0.12	99.89 ± 0.13	99.95 ± 0.09	99.82 ± 0.23	99.60 ± 0.58	**99.95 ± 0.09**
7	77.66 ± 0.90	**80.65 ± 0.59**	89.32 ± 1.42	**90.55 ± 1.88**	89.41 ± 2.38	**91.58 ± 1.42**	93.14 ± 1.79	92.90 ± 2.57	96.14 ± 1.48	95.36 ± 0.56
8	68.92 ± 0.47	**73.24 ± 0.82**	94.69 ± 0.94	94.84 ± 1.57	93.75 ± 0.87	**95.24 ± 0.63**	93.67 ± 2.49	**94.28 ± 0.96**	98.56 ± 0.71	**99.49 ± 0.07**
9	68.62 ± 1.02	**72.85 ± 0.45**	88.96 ± 1.43	**90.36 ± 0.96**	88.30 ± 1.37	**89.72 ± 0.63**	92.45 ± 2.92	91.17 ± 3.04	95.82 ± 2.39	**96.77 ± 0.33**
10	99.69 ± 0.33	99.69 ± 0.33	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00	99.63 ± 0.76	**99.75 ± 0.34**	99.33 ± 0.96	**99.88 ± 0.21**
11	93.44 ± 0.40	**94.97 ± 0.14**	98.88 ± 0.38	**99.07 ± 0.52**	98.91 ± 0.14	**99.14 ± 0.33**	99.55 ± 0.21	**99.89 ± 0.19**	99.60 ± 0.38	**99.62 ± 0.20**
12	91.93 ± 0.86	**93.22 ± 0.60**	100.00 ± 0.00	99.96 ± 0.11	100.00 ± 0.00	100.00 ± 0.00	99.76 ± 0.73	**99.94 ± 0.15**	99.25 ± 0.93	**100.00 ± 0.00**
13	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00	95.56 ± 2.22	**99.54 ± 0.59**
14	92.31 ± 1.94	**93.27 ± 1.32**	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00
15	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00
16	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00	99.31 ± 1.84	97.92 ± 2.69
AA (%)	88.38 ± 0.19	**90.07 ± 0.22**	97.17 ± 0.14	**97.51 ± 0.13**	97.16 ± 0.29	**97.52 ± 0.07**	97.54 ± 0.23	**97.81 ± 0.17**	96.95 ± 2.41	**98.43 ± 0.18**
OA (%)	79.50 ± 0.20	**82.35 ± 0.33**	95.12 ± 0.19	**95.62 ± 0.35**	95.01 ± 0.35	**95.66 ± 0.16**	95.54 ± 0.28	**96.01 ± 0.11**	97.82 ± 0.35	**98.31 ± 0.08**
Kappa × 100	76.83 ± 0.23	**79.98 ± 0.36**	94.37 ± 0.21	**94.96 ± 0.39**	94.27 ± 0.40	**95.06 ± 0.12**	94.90 ± 0.31	**95.43 ± 0.13**	97.52 ± 0.40	**98.08 ± 0.09**

TABLE V

CLASSIFICATION RESULTS BY DIFFERENT METHODS WITH OR WITHOUT SWDIFF FOR THE HOUSTON 2013 DATASET, INCLUDING PER-CLASS ACCURACY, OA, AA, AND KAPPA

	RF		CNN		CAE		SSRN		AMS-M2ESL	
Class No.	Before	After	Before	After	Before	After	Before	After	Before	After
1	82.69 ± 0.07	82.47 ± 0.07	81.14 ± 0.16	**81.44 ± 0.27**	80.88 ± 0.55	**81.58 ± 0.17**	79.68 ± 0.54	77.64 ± 0.87	95.94 ± 3.06	**97.03 ± 0.35**
2	84.40 ± 0.05	**82.89 ± 0.33**	83.95 ± 0.91	83.97 ± 0.64	83.96 ± 1.37	**84.49 ± 0.98**	84.12 ± 1.20	83.04 ± 1.66	98.29 ± 1.55	97.00 ± 0.44
3	41.14 ± 0.91	33.17 ± 1.25	80.12 ± 5.40	80.04 ± 5.54	77.82 ± 1.54	**82.64 ± 1.98**	94.26 ± 0.56	**95.64 ± 1.40**	100.00 ± 0.00	100.00 ± 0.00
4	92.05 ± 0.13	92.42 ± 0.15	92.10 ± 0.64	92.03 ± 0.27	91.60 ± 1.11	**92.58 ± 0.11**	91.43 ± 1.27	90.25 ± 0.67	99.18 ± 0.85	**99.78 ± 0.24**
5	99.86 ± 0.10	99.69 ± 0.18	96.61 ± 1.55	**97.27 ± 0.86**	96.94 ± 0.88	95.77 ± 1.02	99.96 ± 0.06	99.91 ± 0.13	99.86 ± 0.36	**100.00 ± 0.00**
6	94.23 ± 0.72	**94.76 ± 1.18**	83.78 ± 4.35	**91.61 ± 5.70**	85.32 ± 5.55	81.12 ± 0.70	95.11 ± 0.98	**95.80 ± 0.00**	98.94 ± 1.37	**100.00 ± 0.00**
7	86.96 ± 0.29	**87.57 ± 0.17**	87.16 ± 1.12	**88.15 ± 1.19**	87.19 ± 0.69	**87.72 ± 0.67**	83.26 ± 3.23	82.89 ± 9.96	96.83 ± 1.98	**98.69 ± 0.51**
8	46.66 ± 1.87	**51.95 ± 1.52**	66.44 ± 4.78	**68.49 ± 4.07**	63.03 ± 1.60	**68.60 ± 5.18**	75.83 ± 3.56	**87.89 ± 3.96**	99.99 ± 0.03	**100.00 ± 0.00**
9	84.36 ± 0.56	82.53 ± 0.25	86.59 ± 1.65	**86.74 ± 1.08**	85.14 ± 0.80	**86.47 ± 1.01**	85.13 ± 5.54	84.23 ± 3.07	96.52 ± 1.55	**98.30 ± 0.36**
10	34.81 ± 0.24	**38.14 ± 0.35**	56.87 ± 5.61	**62.74 ± 1.54**	59.65 ± 3.14	**60.52 ± 3.01**	51.07 ± 4.09	**59.80 ± 3.89**	97.04 ± 1.49	**99.14 ± 0.66**
11	70.09 ± 0.61	**72.05 ± 0.35**	73.83 ± 1.19	**75.03 ± 0.77**	72.77 ± 1.16	**73.88 ± 0.69**	99.58 ± 0.60	93.12 ± 0.61	98.29 ± 1.36	**99.17 ± 0.39**
12	63.78 ± 1.41	**72.91 ± 0.35**	95.20 ± 2.47	**95.43 ± 0.78**	94.43 ± 0.51	**96.92 ± 1.17**	96.40 ± 2.11	**98.90 ± 0.34**	97.03 ± 1.83	**97.29 ± 0.39**
13	88.68 ± 0.58	88.16 ± 0.67	87.72 ± 2.90	**90.53 ± 1.81**	88.77 ± 0.61	**89.47 ± 1.95**	72.81 ± 0.25	**77.02 ± 2.73**	92.82 ± 3.19	**96.69 ± 1.17**
14	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00	99.92 ± 0.18	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00	98.24 ± 2.27	95.54 ± 0.19
15	73.20 ± 1.30	73.04 ± 0.55	75.52 ± 6.64	**82.92 ± 9.09**	82.24 ± 7.88	74.70 ± 11.27	99.79 ± 0.00	96.62 ± 0.90	100.00 ± 0.00	100.00 ± 0.00
AA (%)	76.20 ± 0.14	**76.78 ± 0.24**	83.13 ± 0.79	**85.09 ± 0.83**	83.32 ± 0.27	**83.76 ± 0.62**	87.23 ± 0.35	**88.18 ± 0.20**	97.93 ± 0.33	**98.57 ± 0.08**
OA (%)	74.32 ± 0.17	**75.42 ± 0.20**	82.22 ± 0.77	**83.64 ± 0.34**	82.05 ± 0.31	**83.04 ± 0.31**	85.83 ± 0.36	**86.82 ± 0.20**	97.91 ± 0.39	**98.62 ± 0.09**
Kappa × 100	72.25 ± 0.18	**73.42 ± 0.22**	80.78 ± 0.84	**82.31 ± 0.38**	80.60 ± 0.34	**81.67 ± 0.34**	84.61 ± 0.38	**85.69 ± 0.23**	97.74 ± 0.42	**98.51 ± 0.10**

TABLE VI

CLASSIFICATION RESULTS BY DIFFERENT METHODS WITH OR WITHOUT SWDIFF FOR THE AUGSBURG DATASET, INCLUDING PER-CLASS ACCURACY, OA, AA, AND KAPPA

	RF		CNN		CAE		SSRN		AMS-M2ESL	
Class No.	Before	After	Before	After	Before	After	Before	After	Before	After
1	88.82 ± 0.29	88.80 ± 0.45	93.94 ± 0.28	**94.32 ± 0.35**	94.64 ± 1.11	94.58 ± 0.51	93.46 ± 2.13	92.89 ± 1.12	96.56 ± 0.44	96.28 ± 0.25
2	97.76 ± 0.06	**98.01 ± 0.07**	98.26 ± 0.34	98.08 ± 0.11	98.35 ± 0.24	97.95 ± 0.33	96.80 ± 0.23	**97.52 ± 0.48**	94.44 ± 0.73	**94.65 ± 0.23**
3	1.68 ± 0.44	**3.73 ± 0.41**	73.41 ± 6.56	**74.34 ± 0.24**	72.80 ± 2.86	**80.84 ± 2.35**	84.41 ± 8.46	82.72 ± 3.06	81.26 ± 2.42	81.01 ± 0.74
4	75.68 ± 0.27	73.78 ± 0.29	84.79 ± 1.23	**86.26 ± 1.01**	86.40 ± 1.40	**87.30 ± 1.00**	91.62 ± 0.60	**91.77 ± 0.92**	96.76 ± 0.80	**97.03 ± 0.11**
5	19.46 ± 1.07	**20.60 ± 0.63**	39.58 ± 1.82	**41.11 ± 2.66**	40.60 ± 3.79	39.20 ± 1.89	49.24 ± 3.38	49.24 ± 0.67	82.97 ± 5.54	**85.24 ± 1.65**
6	0.00 ± 0.00	0.00 ± 0.00	2.24 ± 1.29	1.85 ± 0.14	1.71 ± 0.18	1.59 ± 0.06	10.57 ± 5.44	**16.06 ± 6.05**	63.91 ± 6.18	**71.40 ± 2.48**
7	4.60 ± 0.41	**29.56 ± 1.12**	29.13 ± 10.64	28.56 ± 4.69	21.92 ± 3.11	**24.46 ± 0.48**	48.31 ± 2.06	47.68 ± 0.89	83.02 ± 7.58	82.65 ± 1.49
AA (%)	41.14 ± 0.13	**44.93 ± 0.28**	60.19 ± 1.73	**60.76 ± 1.11**	59.49 ± 0.72	**60.84 ± 0.24**	67.77 ± 0.96	**68.27 ± 1.17**	85.56 ± 0.85	**86.89 ± 0.47**
OA (%)	79.49 ± 0.03	**79.53 ± 0.05**	87.90 ± 0.16	**88.43 ± 0.35**	88.43 ± 0.39	**89.00 ± 0.41**	90.75 ± 0.12	**91.00 ± 0.40**	94.31 ± 0.24	**94.61 ± 0.05**
Kappa × 100	69.12 ± 0.05	**69.31 ± 0.08**	82.60 ± 0.24	**83.40 ± 0.49**	83.35 ± 0.51	**84.23 ± 0.58**	86.78 ± 0.17	**87.11 ± 0.48**	91.79 ± 0.34	**92.23 ± 0.07**

C. Experimental Results

In this section, according to the experimental setting, we employ the proposed method in five different classification methods and discuss the experimental results under the impact of the SWDiff. The details of classification performance are given in Tables IV–VI. The corresponding classification maps obtained by these methods are shown in Figs. 6–10. The results highlighted in boldface represent the better outcomes between utilizing our method and not. The experimental results on the three datasets indicate that, by leveraging the DM's ability to mine prior spectral–spatial information and model spectral–spatial relationships across samples within **HSI**, our proposed SWDiff, successfully enhances the classification performance of both machine learning and DL-based HSIC methods.

1) Indian Pines Dataset: The experimental results on the Indian Pines dataset are shown in Table IV. As can be observed from the tables, the proposed SWDiff has enhanced classification accuracy in most categories. Notably, the largest improvement is seen in the RF method, where OA increased

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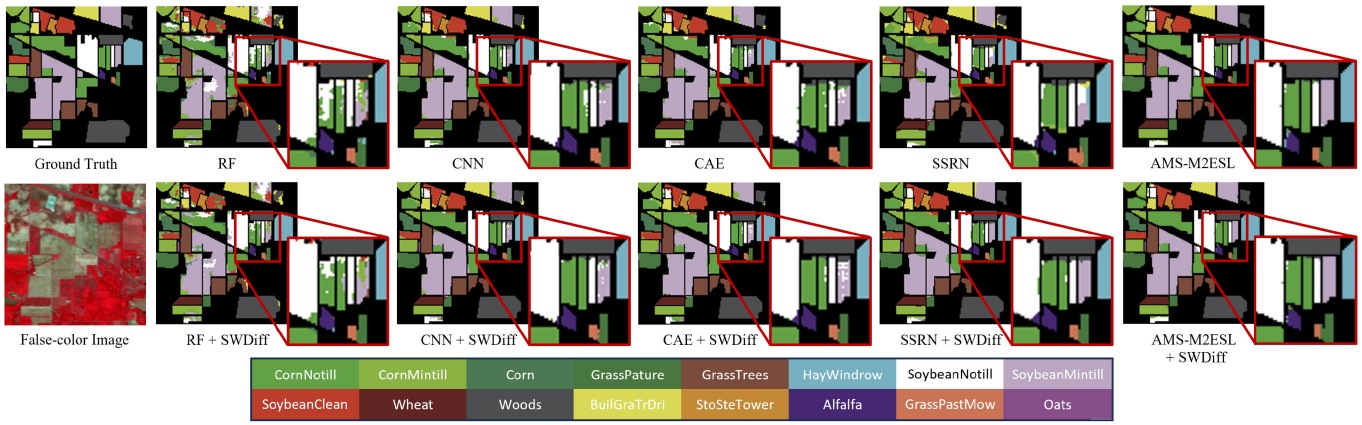


Fig. 6. Ground truth, false-color image, and classification maps obtained by different methods with or without SWDiff for the Indian Pines dataset. The first row is the original performances, whilst the second row is those applied to the SWDiff.

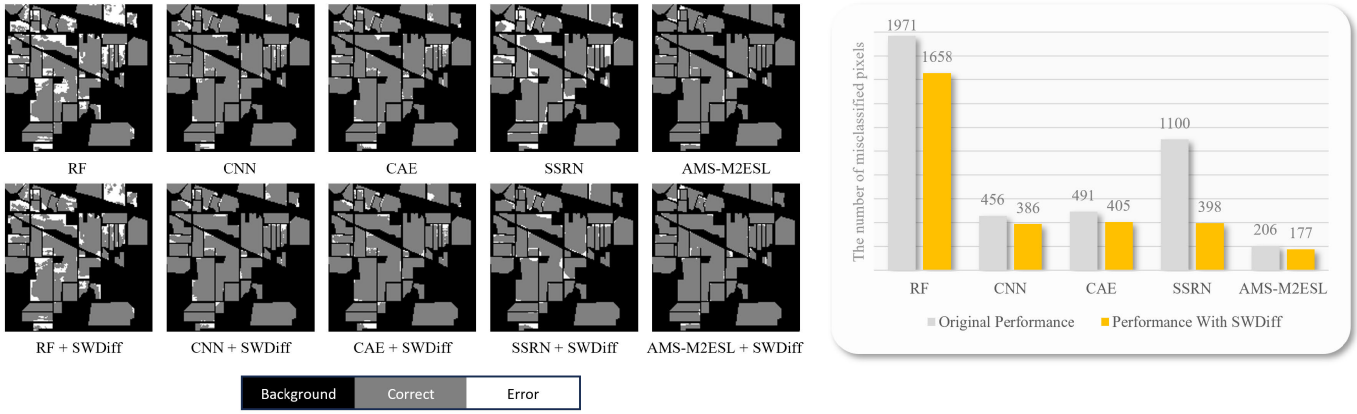


Fig. 7. Left half is error maps obtained by different methods with or without SWDiff for the Indian Pines dataset. The first row is the original performances, whilst the second row is those applied to the SWDiff. Right half is the corresponding statistics of the number of misclassified pixels.

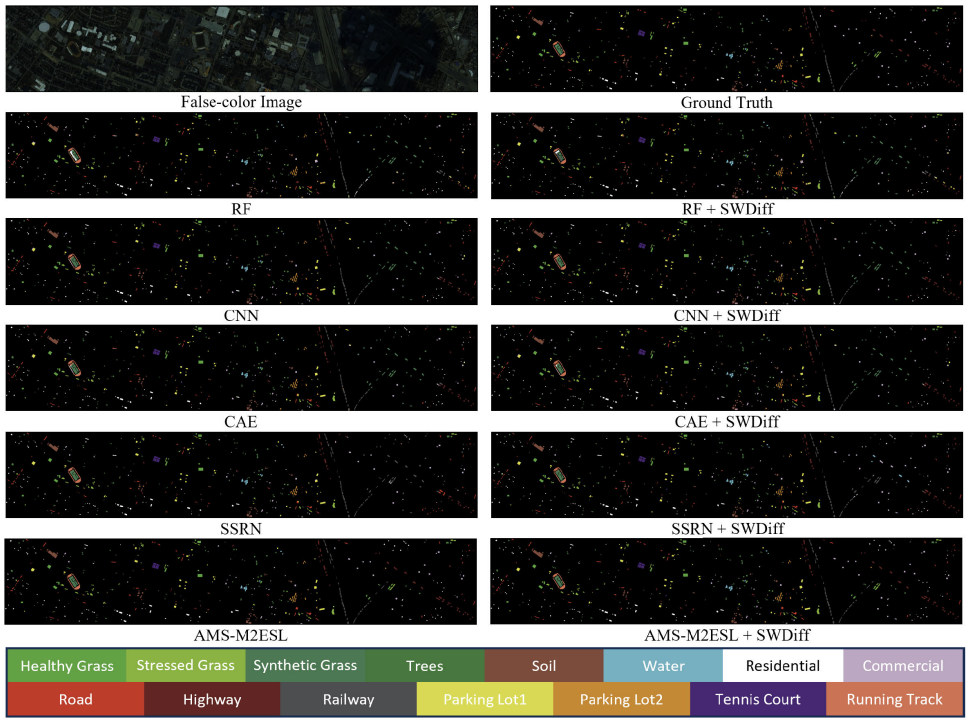


Fig. 8. Ground truth, false-color image, and classification maps obtained by different methods with or without SWDiff for the Houston 2013 dataset. The first column is the original performances, whilst the second column is those applied to the SWDiff.

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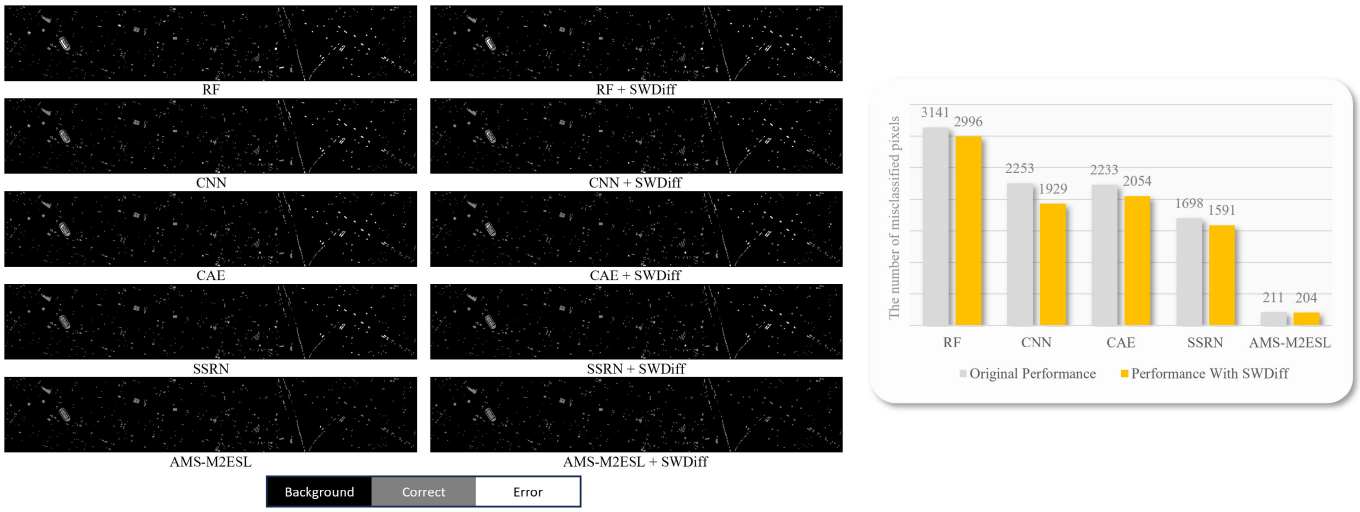


Fig. 9. Left half is error maps obtained by different methods with or without SWDiff for the Houston 2013 dataset. The first row is the original performances, whilst the second row is those applied to the SWDiff. Right half is the corresponding statistics of the number of misclassified pixels.

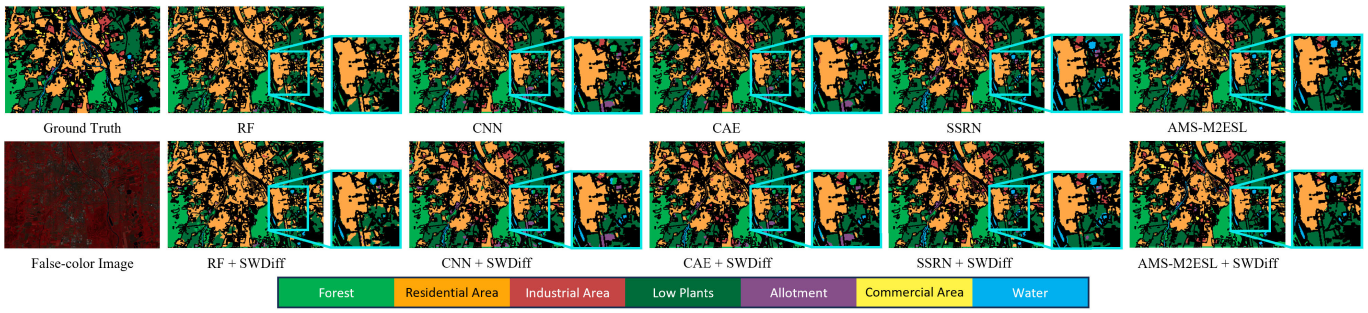


Fig. 10. Ground truth, false-color image, and classification maps obtained by different methods with or without SWDiff for the Augsburg dataset. The first row is the original performances, whilst the second row is those applied to the SWDiff.

from 79.50% to 82.35%. This may be attributed to the fact that, as a machine learning method, RF inherently exhibits weaker classification performance, thus benefiting more distinctly from the enhancements provided by our method. In addition, the four DL algorithms each showed an increase in OA by 0.5%, 0.65%, 0.47%, and 0.49%, respectively. It is important to note that not all categories experienced improved classification accuracy through SWDiff. This could be due to the uncertainty introduced by the DM, as embracing uncertainty is a fundamental aspect of generative models.

The visualized analysis of the results, as shown in Fig. 6, provides an intuitive understanding of the impact of the proposed SWDiff on the original classifiers' outcomes. This impact manifests in a trend where the classifiers' results appear smoother through the effective utilization and mining of hyperspectral prior spatial–spectral information via the DM. This enhances classification accuracy within continuous terrain categories but may reduce the precision at the boundaries between different terrain types.

To demonstrate the effectiveness of the proposed method further, visualization results of error maps and corresponding statistics of misclassified pixels are shown in Fig. 7. From Fig. 7, it is evident that the proposed SWDiff method enhances classification results across all five classifiers, with the most significant improvement observed with the SSRN.

2) Houston 2013 Dataset: The experimental results on the Houston 2013 dataset are shown in Table V. The experimental results demonstrate that our proposed SWDiff enhances the classifiers' performance. A notable deviation from the results obtained on the Indian Pines dataset is observed in the Houston 2013 dataset, where RF, after employing SWDiff, did not achieve a significantly higher accuracy improvement compared to other classifiers. A possible reason could be the greater classification difficulty in the Houston 2013 dataset, specifically, a higher sparsity in data distribution. In addition, the data in Table V underscore the effectiveness of SWDiff. It facilitated improvements in OA for RF, CNN, CAE, SSRN, and AMS-M2ESL by 1.1%, 1.42%, 0.99%, 0.99%, and 0.71%, respectively. Similar to the findings in the Indian Pines dataset, the use of SWDiff in the Houston 2013 dataset might decrease classification accuracy for some categories, potentially due to the randomness introduced by the generative model. However, the outcomes of multiple experimental runs still attest to the efficacy of SWDiff. The visualized classification results and error maps are depicted in Figs. 8 and 9, respectively.

3) Augsburg Dataset: The experimental results on the Augsburg dataset are shown in Table IV. With the application of SWDiff, the OA for RF, CNN, CAE, SSRN, and AMS-M2ESL improved by 0.04%, 0.53%, 0.57%, 0.25%, and 0.3%, respectively. Compared to the first two datasets,

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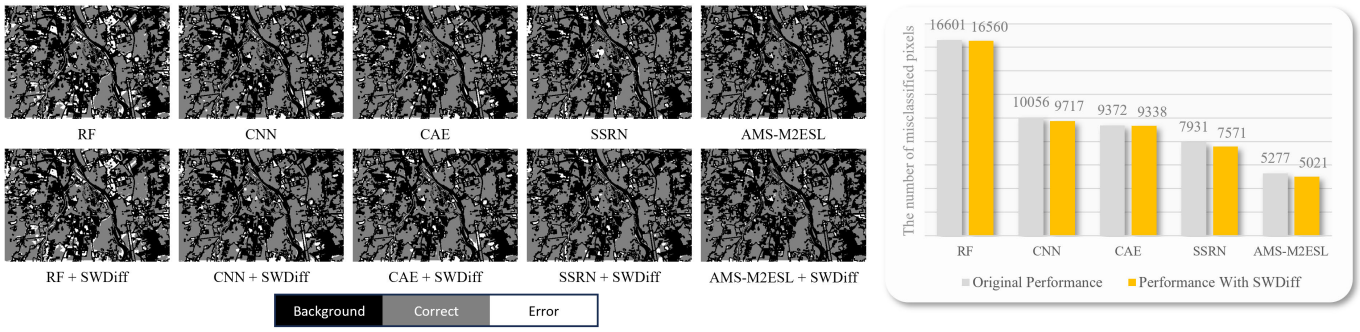


Fig. 11. Left half is error maps obtained by different methods with or without SWDiff for the Augsburg dataset. The first row is the original performances, whilst the second row is those applied to the SWDiff. Right half is the corresponding statistics of the number of misclassified pixels.

the Augsburg dataset exhibited less salient gains in terms of OA when using SWDiff, attributed to its challenges of severely unbalanced labels. However, with the benefit of our SWDiff framework, the insufficient labeled data problem has been alleviated because of complementing the training set by conditional generation in the inference stage. Consequently, regarding AA, SWDiff still made a positive contribution by holding the utilization of spatial–spectral information, with the AA for the aforementioned five methods increasing by 3.79%, 0.57%, 1.35%, 0.5%, and 1.33%, respectively.

The visualized classification results are depicted in Fig. 10. Focusing on the results obtained by the RF method in Fig. 10, it is observable that RF is capable of accurately classifying categories with few labels with the application of the SWDiff framework. For instance, still discuss the results of RF in Fig. 10, within the highlight amplifying area, the water category, which has only 23 training samples, sees a significant improvement in the classification of 1507 test samples. Previously unidentifiable, multiple water bodies are now correctly predicted, which demonstrates the effectiveness of SWDiff and specifically is a piece of strong evidence that our proposed SWDiff could alleviate the challenge of insufficient labeled samples. In the case of SSRN, the classification of the commercial area, which also suffers from this challenge, benefits from the SWDiff.

In addition, Fig. 11 provides further visualization results from the perspective of error maps. These results highlight the improvements in classification results achieved by the SWDiff.

D. Influence of Outset Timestep and Sampling Number

As previously mentioned, hyperparameters related to the DM are eventually crucial to the performance of SWDiff, including the initial timestep t for generation, denoted as t_{outset}, and the number of samples generated for each **HP** during inference stage, denoted as n_{sample}.

The $\mathbf{HP_t}$ contains varied information and noise levels depending on t. Smaller t contains more information but will lose generation diversity, which will decrease the data complement quality. Therefore, there is a tradeoff between more retaining information and better generation diversity. The best performances will be achieved at those suitable balance points between them.

Fig. 12(a)–(c) illustrates the quantitative relationship between the hyperparameters t_{outset}, n_{sample} and metrics

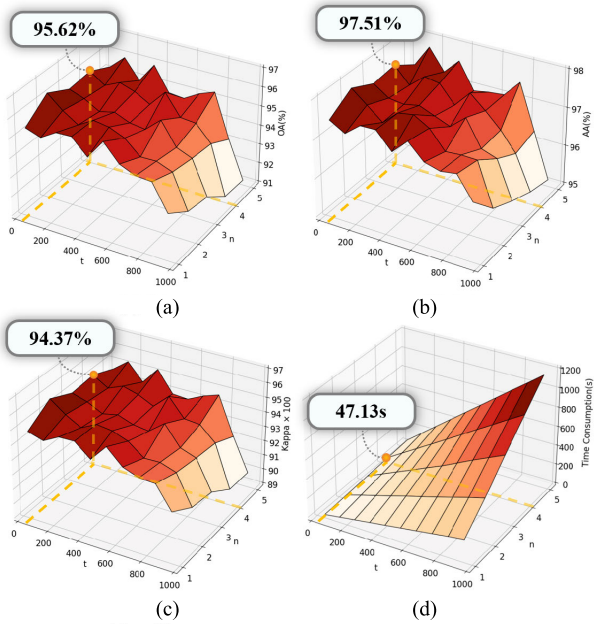


Fig. 12. With deploying SWDiff into CNN on the Indian Pine dataset as an instance, the classification results are changing with varied n and t. The orange point in each subfigure denotes the performance obtained by the specific hyperparameters used in Section IV-C, where n and t are set to 4 and 50, respectively. Deeper color refers to higher accuracy. (a) OA. (b) AA. (c) Kappa. (d) Time consumption.

(including OA, AA, and Kappa) with an instance of deploying SWDiff into CNN on the Indian Pine dataset. A small outset of timesteps t_{outset}, such as 50, retains more spectral and domain information but with limited generation diversity. Conversely, a large t_{outset}, like 800, encompasses excessive noise. While setting large t_{outset} offers greater generation randomness in the inference stage, it can result in a more significant degradation of information in HSIs, ultimately leading to poorer classification accuracy. Therefore, there is a tradeoff between more retaining information and better generation diversity. The best performances will be achieved at suitable balance points between them. Regarding another parameter sampling number n_{sample}, a higher sampling number n_{sample} can improve classification accuracy to some extent but incurs substantial computing consumption. Notably, as n_{sample} increases, the computing consumption correspondingly escalates. To sum up, selecting appropriate hyperparameters for the outset of timesteps t_{outset} and sampling number n_{sample} is crucial to fully unleashing the potential of SWDiff.

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TABLE VII
CLASSIFICATION RESULTS UNDER DIFFERENT NOISE ADDING SCHEDULES ON THE INDIAN PINES DATASET, INCLUDING OA, AA, AND KAPPA

	Schedule	AA (%)	OA (%)	Kappa × 100
RF	Linear	89.98 ± 0.14	82.29 ± 0.16	79.91 ± 0.17
	Cosine	90.07 ± 0.22	82.35 ± 0.33	79.98 ± 0.36
CNN	Linear	97.32 ± 0.13	95.27 ± 0.30	94.57 ± 0.34
	Cosine	97.51 ± 0.13	95.62 ± 0.35	94.96 ± 0.39
CAE	Linear	97.38 ± 0.27	95.51 ± 0.95	94.58 ± 0.57
	Cosine	97.52 ± 0.07	95.66 ± 0.16	95.06 ± 0.12
SSRN	Linear	97.68 ± 0.20	95.89 ± 0.34	95.29 ± 0.39
	Cosine	97.81 ± 0.17	96.01 ± 0.11	95.43 ± 0.13
AMS-M2ESL	Linear	98.27 ± 0.24	98.26 ± 0.13	98.02 ± 0.15
	Cosine	98.43 ± 0.18	98.31 ± 0.08	98.08 ± 0.09

Furthermore, to consider the computing consumption caused by the combination of t_{outset} and n_{sample} quantitively, Fig. 12(d) is drawn based on massive experiments. It is shown from Fig. 12(d) that the time consumption for training the diffusion decoder increases with the increase of the t_{outset} and n_{sample} separately, indicating a positive correlation. Therefore, while seeking the benefits introduced by SWDiff, it is essential to consider the corresponding computational time cost. In other words, a tradeoff between the gains obtained by SWDiff and computational time consumption must be taken into account.

E. Influence of Noise Adding Schedule

The difference between linear and cosine variance schedules relating to adding noise is described theoretically in Section III-A. To compare the influence of these two schedules delivering to our proposed SWDiff quantitatively and verify the superiority of the cosine schedule, we conduct experiments on the Indian Pines dataset amongst varied comparable classification methods. The experimental settings are the same as Section IV-B.

It is shown in Table VII that the performance adapting cosine schedule is overall better than using a linear schedule, denoting the superiority of cosine variance noise adding schedule. Combining with Table VIII, both two schedules can elevate the performance of original methods in the Indian Pines dataset.

F. Ablation Experiments

As previously mentioned, the diffusion decoder of SWDiff utilizes an add operation of embedded image patches of HSI with their corresponding PCA patches, where PCA is regarded as the generative guiding condition. In CV tasks, the selectable conditions of conditional generation are ample. However, most are varied pretrained embedding networks, such as ResNet and FPN, which cannot be directly introduced into our proposed SWDiff. It is because of two reasons. First, there is a considerable gap in the number of image channels between RGB images, where these pretrained networks are trained, and HSIs. Second, some of these pretrained networks own a large number of parameters when compared with the SWDiff decoder, even larger than those of the proposed SWDiff decoder itself. It is conflicted with the lightweight decoder design of SWDiff. Notably, the conditional module in the proposed SWDiff will be called in every iteration of training and inference. So, if these pretrained networks have been adapted in the SWDiff decoder, the time cost would be tremendous. Therefore, the ideal choice of conditional method should be light but efficient, such as PCA and Minimum Noise Fraction (MNF). The MNF transformation is a two-stage process designed to reduce the dimensionality of hyperspectral data while enhancing signal-to-noise ratio [67]. To verify the effectiveness of these PCA components in SWDiff, multiple sets of experiments were conducted on the Indian Pines dataset. The experiments were carried out on five classifiers: RF, CNN, CAE, SSRN, and AMS-M2ESL, with the results presented in Table VIII.

TABLE VIII
CLASSIFICATION RESULTS OF ABLATION EXPERIMENT FOR THE INDIAN PINES DATASET, INCLUDING OA, AA, AND KAPPA

	DM	PCA	MNF	AA (%)	OA (%)	Kappa × 100
RF	✗	✗	✗	88.38 ± 0.19	79.50 ± 0.20	76.83 ± 0.23
	✓	✗	✗	89.93 ± 0.19	82.16 ± 0.19	79.77 ± 0.22
	✓	✗	✓	90.04 ± 0.14	82.31 ± 0.07	79.94 ± 0.07
	✓	✓	✗	90.07 ± 0.22	82.35 ± 0.33	79.98 ± 0.36
CNN	✗	✗	✗	97.22 ± 0.24	95.10 ± 0.41	94.39 ± 0.48
	✓	✗	✗	97.35 ± 0.16	95.40 ± 0.23	94.73 ± 0.27
	✓	✗	✓	97.40 ± 0.10	95.50 ± 0.19	94.83 ± 0.21
	✓	✓	✗	97.51 ± 0.13	95.62 ± 0.35	94.96 ± 0.39
CAE	✗	✗	✗	97.23 ± 0.22	95.15 ± 0.27	94.45 ± 0.31
	✓	✗	✗	97.33 ± 0.14	95.40 ± 0.25	94.73 ± 0.29
	✓	✗	✓	97.36 ± 0.21	95.54 ± 0.42	94.89 ± 0.48
	✓	✓	✗	97.52 ± 0.07	95.66 ± 0.16	95.06 ± 0.12
SSRN	✗	✗	✗	97.54 ± 0.23	95.54 ± 0.28	94.90 ± 0.31
	✓	✗	✗	97.77 ± 0.18	95.73 ± 0.16	95.11 ± 0.19
	✓	✗	✓	97.89 ± 0.13	95.78 ± 0.24	95.17 ± 0.27
	✓	✓	✗	97.81 ± 0.17	96.01 ± 0.11	95.43 ± 0.13
AMS-M2ESL	✗	✗	✗	96.95 ± 2.41	97.82 ± 0.35	97.52 ± 0.40
	✓	✗	✗	98.28 ± 0.19	98.17 ± 0.11	97.91 ± 0.12
	✓	✗	✓	98.40 ± 0.15	98.24 ± 0.10	98.00 ± 0.11
	✓	✓	✗	98.43 ± 0.18	98.31 ± 0.08	98.08 ± 0.09

As indicated in Table VIII, across the five classifiers, when without PCA for generating guidance, the classification results in all three metrics are lower than those obtained with the full implementation of SWDiff, yet still higher than the original performance. Besides, compared with MNF, PCA obtained a better performance overall, except in SSRN, where the MNF-based conditional module received better AA performance than PCA-based. Specifically, when without PCA and utilizing DM only, the OA metrics for RF, CNN, CAE, SSRN, and AMS-M2ESL classifiers are, respectively, 2.66%, 0.3%, 0.25%, 0.19%, and 0.35% higher than the Original performance, and at the same time, 0.19%, 0.22%, 0.26%, 0.28%, and 0.14% lower compared to when utilizing SWDiff framework. When compared with the MNF-based conditional module, the PCA-based module is slightly superior; the OA of PCA-based for five classifiers are 0.04%, 0.12%, 0.12%, 0.23%, and 0.07% higher than MNF-based, respectively.

G. Analysis on Data Augmentation Performance

The proposed SWDiff could be treated as a plug-and-play HSIs data augmentation method. Rotation and flipping are two basic but classical methods for HSIs data augmentation task [64]. Experiments have been conducted to compare the

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TABLE IX

CLASSIFICATION RESULTS UNDER DIFFERENT DATA AUGMENTATION METHODS, INCLUDING FLIPPING, ROTATION, AND SWDIFF. THE OPTIMAL AND SUBOPTIMAL CLASSIFICATION PERFORMANCE IN EACH N IS DENOTED AS BOLD AND UNDERLINED, RESPECTIVELY

N	Augmentation Method	AA (%)	OA (%)	Kappa × 100
0	None	83.13 ± 0.79	82.22 ± 0.77	80.78 ± 0.84
1	Flip	84.94 ± 0.55	82.90 ± 0.49	81.51 ± 0.52
	Rotate	<u>85.03 ± 0.47</u>	<u>83.20 ± 0.32</u>	<u>81.84 ± 0.35</u>
	SWDiff	**85.09 ± 0.83**	**83.64 ± 0.34**	**82.31 + 0.38**
2	SWDiff + Flip	<u>85.31 ± 0.53</u>	<u>83.74 ± 0.51</u>	<u>82.42 ± 0.55</u>
	SWDiff + Rotate	**85.85 ± 0.45**	**83.92 ± 0.36**	**82.62 ± 0.40**

TABLE X

NUMBER OF NETWORK PARAMETERS AND MACS COMPARISON AMONGST DDPM, IDDPM, U-VIT, AND PROPOSED SWDIFF

Decoder	N_Params (M)	MACs (G)
DDPM	35.97	1.58
64 ch IDDPM	9.79	0.43
96 ch IDDPM	21.91	0.95
128 ch IDDPM	38.86	1.67
192 ch IDDPM	87.21	3.70
U-ViT-S	45.45	3.04
U-ViT-M	135.75	2.41
U-ViT-L	287.68	18.78
U-ViT-H	507.46	8.62
SWDiff	**1.89**	**0.30**

data augmentation performance of SWDiff with other basic methods. Specifically, CNN was selected as the classifier, and the dataset chosen was Houston 2013. The angle of rotation is set as 90° while the direction of flipping is set as horizontal. The SWDiff settings followed those detailed in Section IV-B. Depending on the number of data augmentation operations, N, the experimental results were divided into operations performed once and twice, i.e., $N = 1$ and $N = 2$. The experimental results are presented in Table IX.

The results from Table IX demonstrate that SWDiff achieves better final classification performance compared to both flipping and rotation. Specifically, when only a single data augmentation operation is employed, the proposed SWDiff yields superior classification outcomes relative to flipping and rotation. Furthermore, when other data augmentation methods are used in conjunction with SWDiff, a dual augmentation approach further enhances classification performance, illustrating the plug-and-play characteristic of SWDiff.

H. Analysis on Parameters Amount and MACs of DMs Decoder

U-Net is primarily chosen as the decoder of DMs. This kind of decoder comprises both transformer and convolutional layers, as exemplified in DDPM [28] and IDDPM [30], or transformer layers only, such as U-Vit [68]. Put it another way, there are fewer works based on DMs employing solely convolutional layers in their decoders. In contrast, the proposed SWDiff innovatively attempts to use only convolutional layers in the decoder's design, with the network structure illustrated in the previous Fig. 2. This design not only retains the effectiveness of SWDiff but also reduces the number of network parameters extremely compared to methods like DDPM.

The quantitative differences amongst these decoder networks in the number of parameters and MACs are presented in Table X. Here, MAC refers to multiply-accumulate operation, which is roughly equal to FLOPs (floating point operations) times 2. Specifically, MACs are a fundamental metric for quantifying network computational complexity. Each MAC consists of one multiplication followed by one addition, and the total number of MACs directly reflects the network's processing workload. Accurately measuring MACs is essential for evaluating the efficiency and performance of different network architectures. To compare under fair conditions, the size of input among all these decoder networks is set to the same number, 16, which is in line with the hyperparameter in our SWDiff experiments. In addition, to fit the scenario of HSI, the number of first network layer channels is set as 220 within all decoder networks to be compared, equalling the situation of the Indian Pines dataset. The rest of the settings of the decoder are the same as the corresponding papers. Consequently, the experimental results might be different from those published in their own papers to some degree.

Table X reveals that the proposed SWDiff decoder is lighter than the key milestones in the DM family. Specifically, the parameter amount of the proposed SWDiff decoder, 1.89M, is 19.03 times less than the decoders of DDPM; compared with IDDPMs [30], it is 5.18, 11.59, 20.56, and 46.14 times less than the decoders of 64 ch IDDPM, 96 ch IDDPM, 128 ch IDDPM, and 192 IDDPM, respectively; In contrast to U-ViTs [68], it is 24.05, 71.83, 152.21, and 268.50 times less than the decoders of U-ViT-S, U-ViT-M, U-ViT-L, and U-ViT-H, respectively. Regarding MACs, the proposed SWDiff requires only 0.3G MACs, which is 1.28G fewer than the DDPM's decoder. Compared to IDDPM decoders with 64, 96, 128, and 192 channels, the SWDiff decoder's MACs is fewer by 0.13G, 0.65G, 1.37G, and 3.40G, respectively. The lightweight design of the SWDiff decoder is superior especially when compared with the solely transformer-based U-ViT decoders. Specifically, the SWDiff's MACs is lower by 2.74, 2.11, 18.48, and 8.32 compared to the U-ViT-S, U-ViT-M, U-ViT-L, and U-ViT-H decoders, respectively.

I. Analysis on Time Consumption of Inference Stage

In Section III-B, DDIM was discussed on the keyword of the sampling speed. To display the time consumption difference between our proposed SWDiff employing the DDIM accelerating sample algorithm or not, broad experiments have been conducted. Specifically, the experiments involve the time consumption of all five comparing methods on three datasets (Indian Pines, Houston 2013, and Augsburg). In terms of the settings of DDIM, the most crucial hyperparameter is the length of the reconstructed sequence which is a subset of the original forwarding processing $\mathbf{HP}_{1:T}$. This length is denoted as S and set as 6 in our experiments. For example, when the t_{outset} is 50 and 150, the equalling reconstructed sequence is $\{\mathbf{HP}_1, \mathbf{HP}_{10}, \mathbf{HP}_{20}, \mathbf{HP}_{30}, \mathbf{HP}_{40}, \mathbf{HP}_{50}\}$ and $\{\mathbf{HP}_1, \mathbf{HP}_{30}, \mathbf{HP}_{60}, \mathbf{HP}_{90}, \mathbf{HP}_{120}, \mathbf{HP}_{150}\}$,

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TABLE XI

WITH EMPLOYING OUR PROPOSED SWDIFF, THE TIME CONSUMPTION RESULTS OF USING DDIM OR NOT. COMPARING METHODS INCLUDE RF, CNN, CAE, SSRN, AND AMS-M2ESL. DATASETS INVOLVE THE INDIAN PINES, HOUSTON 2013 DATASET, AND AUGSBURG

Dataset	Inference Algorithm	Inference (Sampling) Time Consumption (s)				
		RF	CNN	CAE	SSRN	AMS-M2ESL
Indian Pine	DDPM	17.88 ± 0.35	47.13 ± 0.35	35.13 ± 0.35	176.88 ± 0.83	7.50 ± 1.85
	DDIM	4.50 ± 0.53	6.0 ± 0.53	4.50 ± 0.53	7.25 ± 0.46	2.00 ± 0.93
Houston 2013	DDPM	114.88 ± 0.35	114.75 ± 0.46	404.25 ± 0.46	90.50 ± 0.53	7.14 ± 1.68
	DDIM	28.13 ± 0.35	28.50 ± 0.53	17.25 ± 0.46	11.75 ± 0.46	2.00 ± 1.00
Augsburg	DDPM	18.75 ± 0.46	62.13 ± 0.35	124.25 ± 0.46	49.13 ± 0.35	7.29 ± 1.89
	DDIM	1.63 ± 0.52	7.63 ± 0.52	7.63 ± 0.52	3.50 ± 0.53	1.71 ± 0.49

TABLE XII

SETTINGS OF HYPERPARAMETERS t_{outset} AND n_{sample} IN COMPARATIVE EXPERIMENTS ON THE INDIAN PINES, HOUSTON 2013, AND AUGSBURG DATASETS, RESPECTIVELY

Dataset	Hyperparameter	RF	CNN	CAE	SSRN	AMS-M2ESL
Indian Pines	t_{outset}	25	50	50	150	25
	n_{sample}	3	4	3	5	1
Houston 2013	t_{outset}	25	25	150	50	25
	n_{sample}	5	5	3	2	1
Augsburg	t_{outset}	75	50	100	100	25
	n_{sample}	1	5	5	2	1

TABLE XIII

CLASSIFICATION RESULTS UNDER DIFFERENT INFERENCE SCHEDULES ON THE INDIAN PINES DATASET, INCLUDING OA, AA, AND KAPPA

	Schedule	AA (%)	OA (%)	Kappa × 100
RF	DDIM	90.03 ± 0.12	82.25 ± 0.10	79.85 ± 0.12
	DDPM	90.07 ± 0.22	82.35 ± 0.33	79.98 ± 0.36
CNN	DDIM	97.43 ± 0.06	95.59 ± 0.22	94.95 ± 0.25
	DDPM	97.51 ± 0.13	95.62 ± 0.35	94.96 ± 0.39
CAE	DDIM	97.42 ± 0.11	95.64 ± 0.30	95.00 ± 0.21
	DDPM	97.52 ± 0.07	95.66 ± 0.16	95.06 ± 0.12
SSRN	DDIM	97.79 ± 0.12	95.71 ± 0.17	95.08 ± 0.25
	DDPM	97.81 ± 0.17	96.01 ± 0.11	95.43 ± 0.13
AMS-M2ESL	DDIM	98.41 ± 0.06	98.22 ± 0.10	97.98 ± 0.12
	DDPM	98.43 ± 0.18	98.31 ± 0.08	98.08 ± 0.09

respectively. To obtain the time consumption associated with the classification results of Table IV–VI, the hyperparameters of t_{outset} and n_{sample} in this Section IV-H are set as same to Section IV-C. The settings and the time consumption experimental results are detailed below in Tables XI and XII.

The experimental results show that DDIM could reduce the time consumption. Especially, when t_{outset} is large such as 150 of SSRN on the Indian Pines dataset and CAE on the Houston 2013 dataset, DDIM accelerates inference speed dramatically. This is because the stride of the reconstructed sequence is different due to the t_{outset}. Larger t_{outset} will lead to a larger stride under the same setting of S, eventually leading to sharper acceleration. However, deploying DDIM will lead to slight classification degradation, as shown in Table XIII. Therefore, there is a tradeoff between faster sampling speed enabled by the DDIM algorithm and higher classification accuracy. Thanks to the lightweight decoder design in our scenario to some degree, we find that even the maximum average inference time consumption without employing the DDIM algorithm, 404.25 s, is less than 7 min, which could be considered to be acceptable in the HSIC task. So, in this tradeoff question, we choose higher accuracy with adequate sampling speed via a lightweight decoder design. On the other hand, if there is a specific requirement for the inference time of the proposed SWDiff method with a slight margin of accuracy deterioration, the DDIM algorithm can be applied.

V. CONCLUSION

In HSIC tasks, inadequate utilization of spatial–spectral information and insufficient labeled samples are two intrinsic challenges. The current methods to tackle them confront hard to modeling the spatial–spectral relationship and suffer from unstable training, respectively. To address these issues, we proposed SWDiff, a novel approach for HSIC that addresses both these two challenges. To fully utilize spatial–spectral information of HSIs, SWDiff leverages the inherent strengths of DMs to model the underlying spatial–spectral information. To alleviate insufficient labeled samples, SWDiff generates highly representative and diverse training samples via the well-trained decoder, significantly enhancing classifier performance. What's more, the lightweight design of the proposed decoder aligns with the need for efficient computational performance, making SWDiff suitable for deployment in scenarios with computational constraints. Our comprehensive experiments across three standard datasets (Indian Pines, Houston 2013, and Augsburg) demonstrate that SWDiff effectively improves classification accuracy.

Looking forward, we aim to explore the integration of DMs with other remote sensing tasks, from discriminative tasks such as segmentation to generative tasks like image translation. By continuing to refine the proposed SWDiff, we hope to contribute to the advancement of geospatial data analysis technologies.

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